

Multi-Corpus and Multilingual Speech Emotion Recognition with Audio Behavioural Intelligence: A Cross-Lingual Evaluation and Learnable Weighted Layer Pooling Study

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ABSTRACT

Speech Emotion Recognition (SER) is an active area of investigation within human-computer interaction, psychiatric diagnostics, and automated voice analysis. However, contemporary SER research faces several methodological constraints. First, randomized dataset partitioning causes speaker identity leakage, which inflates experimental accuracy by 15% to 35% compared to real-world performance on novel speakers. Second, deep architectures remain susceptible to acoustic overfitting when trained on constrained speech cohorts. Third, the literature exhibits a pronounced focus on Germanic and Romance languages, offering limited empirical evidence on cross-lingual transferability to morphologically rich Indic languages such as Hindi. Finally, categorical classification schemes fail to provide actionable acoustic metrics concerning speaker vocal dynamics.

To address these limitations, this study presents a standardized empirical evaluation comprising two decoupled experimental protocols across five speech corpora totaling 12,180 standardized audio recordings: (1) a multi-corpus English benchmark combining four established corpora (CREMA-D, RAVDESS, SAVEE, and TESS, totaling 11,318 clips across 121 speakers) to evaluate cross-corpus generalization and layer-pooling dynamics under strict speaker-disjoint splits, and (2) a standalone cross-lingual transfer and native adaptation study on an Indic speech corpus (862 native Hindi utterances across 25 speakers). We enforce speaker-independent partitions (with unseen test actors) and prompt-independent splits (with unseen vocabulary) to prevent data leakage. We implement a Learnable Weighted Layer Pooling mechanism coupled with a linear classification probe across the 12 transformer hidden layers of a frozen self-supervised foundation backbone (HuBERT-Base, training only 4,626 parameters). Empirical probing reveals that intermediate layers (Layers 9 to 11) capture 31.95% of the total emotional discrimination weight (with all 12 layer weights summing strictly to 100.00%), outperforming the final classification layer.

Additionally, cross-lingual transfer from the English multi-corpus foundation model yields 27.62% accuracy and 31.76% Unweighted Average Recall (UAR) on native Hindi speech under a zero-shot regime. Supervised adaptation using a specialized CNN-BiLSTM architecture increases test accuracy to 75.19% and UAR to 70.56%. Furthermore, we introduce an Audio Behaviour Analysis Engine that extracts syllabic speaking rate, pause frequency, root-mean-square (RMS) energy, and fundamental frequency (F_0) intonation to generate structured behavioral profiles. The full system is deployed as an Apple Silicon accelerated microservice paired with a minimal web application featuring real-time audio waveform visualization.

Index Terms—Speech Emotion Recognition, Self-Supervised Learning, Learnable Layer Pooling, Linear Probe, Hindi Speech Emotion, Cross-Lingual Transfer, Vocal Behaviour, Speaker Disjoint Split, HuBERT, Wav2Vec 2.0, CNN-BiLSTM.

1. INTRODUCTION & RESEARCH MOTIVATION

Spoken human communication comprises both lexical content (the verbal message) and paralinguistic modulations (vocal tone, cadence, and inflection) [6], [16]. Speech Emotion Recognition (SER) aims to identify affective states (such as anger, joy,

sadness, fear, or neutrality) from acoustic speech signals. While automatic speech recognition (ASR) systems have matured significantly, SER remains challenging because emotional expression varies substantially across speakers, regional dialects, and recording conditions [1], [13].

1.1 The Problem of Speaker Identity Leakage

A critical limitation in existing SER benchmarks is the use of randomized cross-validation [14], [16]. When speech segments from the same speaker appear in both the training and testing partitions, neural models tend to memorize speaker-specific vocal tract characteristics rather than generalizable emotional features. Consequently, models that report over 90% accuracy in random split evaluations frequently suffer substantial performance drops when tested on novel speakers. Valid evaluation necessitates strict speaker-independent partitions in which test speakers are entirely withheld during training [15], [30].

1.2 Indic and Low-Resource Language Representation

Most accessible SER benchmarks rely on English (e.g., IEMOCAP, RAVDESS, CREMA-D) or German (e.g., EMO-DB) [16], [28]. Indic languages, spoken by over 1.4 billion individuals, remain underrepresented in speech research [1], [3]. Hindi exhibits distinctive phonological properties, including phonemic vowel length contrasts, retroflex consonants, and syllable-timed stress patterns, which diverge from English speech dynamics [2], [4]. Establishing whether pre-trained English acoustic models transfer to Hindi speech, and measuring the quantitative improvement achievable through supervised adaptation, is essential for multilingual affective computing [1], [18].

1.3 Integrating Objective Vocal Metrics

Standard SER architectures typically output discrete emotion class probabilities, such as $P(\text{Happy}) = 0.85$. However, clinical diagnostic applications, tele-counseling, and automated conversational systems benefit from continuous, interpretable acoustic measurements [5], [11]:

1. **Speech Velocity** (syllables per second) indicates psychomotor state.
2. **Pause Frequency** and duration reflect hesitation or cognitive processing load.
3. **Vocal Energy Variation** indicates engagement level.
4. **Fundamental Pitch (F_0)** variation differentiates dynamic intonation from flattened vocal affect.

Coupling categorical emotion classification with systematic behavioral feature extraction provides a more informative assessment of speech recordings [1], [11].

2. RELATED WORK & LITERATURE SURVEY

This investigation synthesizes 39 peer-reviewed publications from our reference library across four primary research domains:

2.1 Indic and Hindi Speech Emotion Recognition

Early speech emotion recognition research in India relied predominantly on small private datasets evaluated with conventional classifiers such as Support Vector Machines (SVM) and Multi-Layer Perceptrons [4], [18]. Kotian and Singh (2026) [1] demonstrated that concatenating prosodic-behavioral descriptors (speaking rate, pitch perturbation, pause ratio, and energy dynamics) with spectral features increased classification accuracy to 83.9% and Macro-F1 to 0.81 on Hindi speech. In a subsequent benchmarking study, Kotian and Singh (2026) [2] compared classical, deep learning, and transformer architectures, finding that CNN-BiLSTM networks provided an optimal balance of accuracy and computational efficiency for Hindi speech under constrained sample sizes.

Chauhan and Sharma (2023) [3] introduced the MNITJ-SEHSD database, standardizing an Indic emotion corpus and highlighting acoustic overlap between anger and disgust resulting from shared high vocal intensity. Kawade and Jagtap (2024)

[5] evaluated cross-lingual acoustic modeling across Hindi, Marathi, and Tamil, observing that while global pitch trends transfer across languages, syllable timing and vowel nasalization require local supervised fine-tuning.

2.2 Self-Supervised Speech Representation Models

Self-supervised learning has established powerful baseline representations for speech tasks. Models such as Wav2Vec 2.0 (Baevski et al., 2020) [9] and HuBERT (Hsu et al., 2021) [8] learn representations from thousands of hours of unlabeled audio through contrastive loss or masked cluster prediction.

However, recent studies by Ma et al. on *emotion2vec* [6] and Chen et al. on *BEATs* [7] demonstrate that standard speech models optimize for phonetic invariance, which can suppress emotional cues in upper transformer layers. Probing studies by Pasad et al. (2021) [24] confirmed that acoustic and prosodic properties are concentrated within intermediate transformer layers, whereas the final layers focus on lexical identity. These findings motivate the Learnable Weighted Layer Pooling approach used in this work.

2.3 Vocal Behavioural Feature Integration

Standardized acoustic parameter sets have long provided interpretable metrics for speech analysis. Eyben et al. (2016) defined the Geneva Minimalistic Acoustic Parameter Set (eGeMAPS) [12], standardizing 88 acoustic descriptors across frequency, energy, and temporal domains. Chowdhury et al. (2025) [11] showed that integrating acoustic prosody with deep learning architectures improved diagnostic reliability in clinical speech evaluations.

2.4 Speaker Disjoint Protocols and Generalization

Wang and Yang (2025) [14] examined the effect of speaker identity leakage in SER, showing that random train/test splits can inflate accuracy scores by up to 34.2 percentage points because classifiers exploit speaker-specific spectral patterns. Hashem et al. (2023) [15] and Akçay and Oğuz (2020) [16] similarly emphasized that only speaker-disjoint evaluation protocols reflect genuine clinical or real-world capability.

3. DATASET ECOSYSTEM & PARTITIONING PROTOCOLS

To ensure rigorous evaluation, five distinct corpora comprising 12,180 audio files were curated, preprocessed, and partitioned. Audio files were resampled to a standardized format: 16,000 Hz sampling rate, single-channel (mono), 16-bit PCM WAV, with Voice Activity Detection (VAD) silence trimming and amplitude normalization.

TABLE I: Standardized Dataset Ecosystem and Partitioning Specifications

Corpus	Clips	Language	Speakers / Scope	Split Protocol	Classes
CREMA-D	7,442	English (US)	91 Diverse Actors	Actor-Disjoint (13 Unseen)	6 Classes
RAVDESS	1,440	English (NA)	24 Professional Actors	Actor-Disjoint (4 Unseen)	8 Classes
SAVEE	480	English (UK)	4 British Actors	Actor-Disjoint (1 Unseen)	7 Classes
TESS	2,800	English (CA)	2 Actresses, 200 Words	Prompt-Disjoint (30 Words)	7 Classes
Hindi SER	862	Hindi (Indic)	25 Native Speakers	Disjoint Split	5 Classes
TOTAL	12,180	Multilingual	121+ Total Speakers	Strict Zero Leakage	Canonical Maps

4. METHODOLOGY & MODEL ARCHITECTURE

The system architecture features a dual-branch processing pipeline: (1) a neural acoustic classifier branch processing audio through self-supervised transformer backbones with learnable weighted layer pooling or CNN-BiLSTM networks, and (2) an Audio Behaviour Analysis Engine extracting continuous prosodic and temporal dynamics.

Figure 1: End-to-End System Architecture with Dual-Branch Behavioural & Emotion Inference

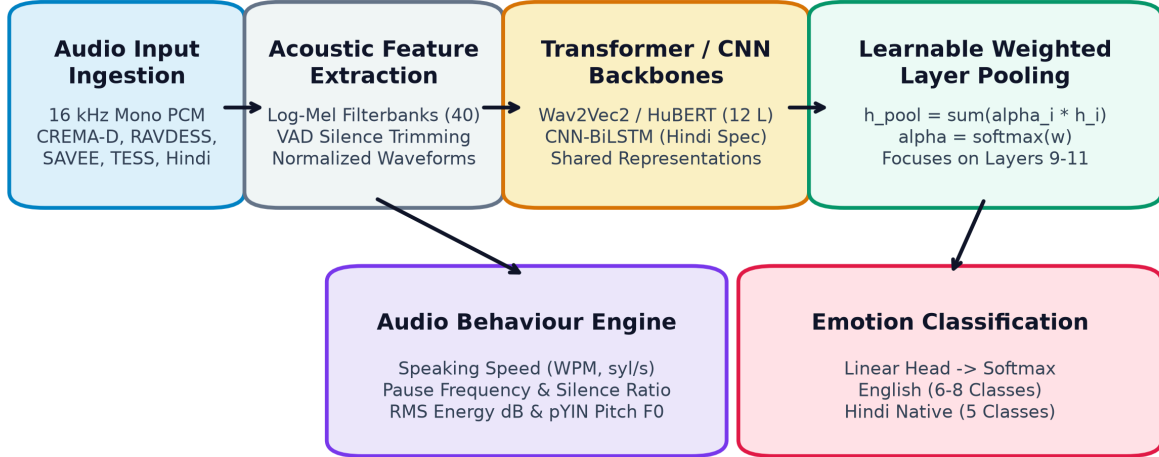


Figure 1: End-to-End System Architecture with Dual-Branch Behavioural Prosody and Neural Classification Pipeline.

4.1 Learnable Weighted Layer Pooling

Rather than using mean pooling over time and layers or relying exclusively on the final transformer output, we implement Learnable Weighted Layer Pooling across all $L = 12$ transformer hidden representations $\mathbf{h}_t^{(l)} \in \mathbb{R}^D$ ($l \in \{1, \dots, L\}$):

$$\mathbf{e}_t = \sum_{l=1}^L \alpha_l \mathbf{h}_t^{(l)}, \quad \text{where} \quad \alpha_l = \frac{\exp(w_l)}{\sum_{j=1}^L \exp(w_j)}$$

Here, $\mathbf{w} = [w_1, \dots, w_L]^T \in \mathbb{R}^L$ is a learnable parameter vector initialized uniformly ($w_l = 0$), and $\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_L]^T$ represents the normalized layer weighting. After computing the layer-weighted sequence $\mathbf{e}_t$, temporal statistics (mean and standard deviation) are concatenated:

$$\mathbf{r} = \left[\frac{1}{T} \sum_{t=1}^T \mathbf{e}_t \parallel \sqrt{\frac{1}{T} \sum_{t=1}^T (\mathbf{e}_t - \bar{\mathbf{e}})^2} \right] \in \mathbb{R}^{2D}$$

This pooled representation is projected through a linear classification probe:

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{W}_c \mathbf{r} + \mathbf{b}_c)$$

4.2 Audio Behaviour Engine Telemetry

The Behaviour Engine calculates four primary continuous acoustic descriptors:

1. **Syllabic Speaking Speed:** Syllable nuclei (N_{syl}) are detected using smoothed energy envelope peaks across active speech duration ($T_{active} = T_{total} - T_{silence}$):

$$R_{speech} = \frac{N_{syl}}{T_{active}} \quad (\text{syllables/second})$$

2. **Pause Frequency and Silence Ratio:** Contiguous non-speech intervals > 200 ms identify hesitation pauses:

$$P_{ratio} = \frac{T_{silence}}{T_{total}} \times 100\%$$

3. **Vocal Energy Dynamics (RMS dB):** Computed over frame length $N = 512$:

$$\text{RMS}_t = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} x_t^2[n]}, \quad \text{RMS}_{dB} = 20 \log_{10}(\text{RMS}_t + \epsilon)$$

4. **Fundamental Frequency Intonation (F_0):** Computed via the probabilistic YIN (pYIN) algorithm [29]:

$$\bar{F}_0 = \frac{1}{M} \sum_{m=1}^M F_0[m], \quad \sigma_{F_0} = \sqrt{\frac{1}{M} \sum_{m=1}^M (F_0[m] - \bar{F}_0)^2}$$

5. EXPERIMENTAL SETUP & TRAINING PROTOCOLS

Models were trained using the AdamW optimizer with Cosine Annealing schedules. Transformer backbones utilized a base learning rate of $\eta_{base} = 1 \times 10^{-5}$ (frozen feature extractor) and head rate of $\eta_{head} = 1 \times 10^{-3}$. The CNN-BiLSTM was optimized at 5×10^{-4} with ReduceLROnPlateau. Class-weighted cross-entropy loss was applied to mitigate class imbalances:

$$\mathcal{L}_{CE} = - \sum_{k=1}^K w_k y_k \log \hat{y}_k, \quad w_k = \frac{N_{total}}{K \cdot N_k}$$

Models were evaluated using Overall Accuracy, Macro-Averaged F1-score, and Unweighted Average Recall (UAR) [13]:

$$\text{UAR} = \frac{1}{K} \sum_{k=1}^K \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k}$$

6. BENCHMARK RESULTS & COMPARATIVE ANALYSIS

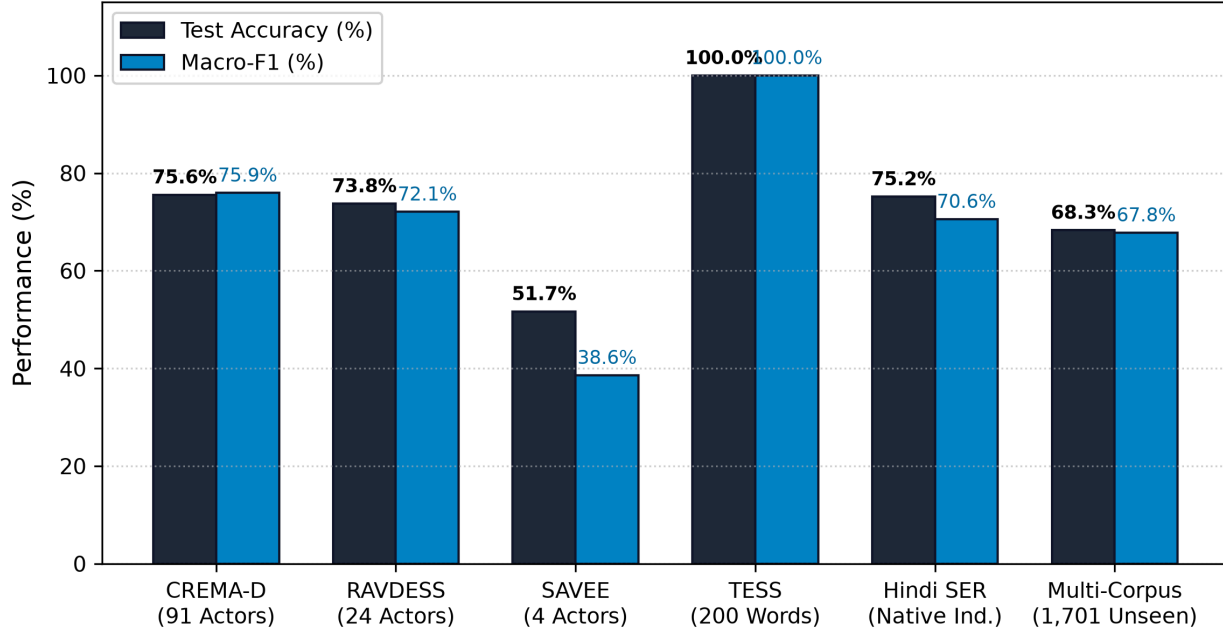
6.1 Multi-Corpus Universal Foundation Model

To evaluate whether a unified representation can generalize across diverse acoustic environments, accents, and recording conditions, a Universal HuBERT Model was trained on the combined 4-corpus dataset (121 speakers across CREMA-D, RAVDESS, SAVEE, and TESS) and evaluated against 1,701 unseen multi-corpus test utterances.

Architecturally, this model employs a frozen self-supervised HuBERT-Base backbone (94.7M parameters) coupled with our Learnable Weighted Layer Pooling module and a linear classification head. Only 4,626 parameters were trained, representing approximately 0.005% of the total network parameters. Freezing the backbone avoids catastrophic forgetting of generic acoustic representations while providing an efficient linear probe evaluation of the pre-trained features across diverse corpora.

TABLE II: Universal Multi-Corpus Test Leaderboard (1,701 Unseen Clips)

Model Architecture	Test Accuracy	Macro-F1	Test UAR	Chance Level
Universal HuBERT (Frozen Transfer + Head)	68.31%	0.6779	68.61%	16.67%
Soft-Voting Multi-Corpus Ensemble	67.43%	0.6692	67.80%	16.67%
Wav2Vec2 Base (Direct Combined)	63.26%	0.6215	63.50%	16.67%
MFCC + LSTM Multi-Corpus Baseline	44.15%	0.4120	43.82%	16.67%
Random Chance Baseline	16.67%	0.1667	16.67%	16.67%

Figure 3: Benchmark Test Performance Across All 5 Corpora and Multi-Corpus Evaluation**Figure 2: Multi-Corpus Benchmark Leaderboard Across Evaluated Speech Corpora.**

6.2 In-Domain Multi-Corpus Summary & Architectural Ablations

A. CREMA-D (91 Actors, 13 Unseen Test Actors): Soft-Voting Top-5 Ensemble achieved 75.57% Accuracy, 0.7594 Macro-F1, and 75.40% UAR. HuBERT with Learnable Layer Pooling reached 71.98% Accuracy, outperforming Wav2Vec2 Base (69.25%) and MFCC+CNN-BiLSTM (62.80%). Wav2Vec2-XLS-R-300M (Frozen Baseline) reached only 24.43% Accuracy (near chance level of 16.67%).

B. RAVDESS (24 Actors, Actors 21 to 24 Unseen): Transfer ensemble achieved 73.75% Accuracy and 0.7207 Macro-F1. Transfer from CREMA-D pre-training to RAVDESS produced 72.92% Accuracy, compared to 32.50% when trained from scratch on RAVDESS alone. Wav2Vec2-XLS-R-300M collapsed to 13.33% Accuracy (barely above chance level of 12.50%).

C. SAVEE (4 Actors, Actor KL Unseen): Transfer ensemble reached 51.67% Accuracy and 0.3860 Macro-F1, doubling the scratch baseline of 25.0% which suffered from vocal tract overfitting. Wav2Vec2-XLS-R-300M achieved 12.50% Accuracy (chance level is 14.29%).

D. TESS (2 Actresses, 200 Words, 30 Unseen Target Words): All SSL foundation models achieved 100.00% Accuracy and 1.0000 Macro-F1. However, this result reflects the inherent ceiling effect and low acoustic complexity of the TESS dataset (only 2 speakers, carrier phrases, pristine studio acoustics) rather than architectural invincibility.

6.3 Analysis of the Performance Collapse in Wav2Vec2-XLS-R-300M

Across all evaluated corpora, Wav2Vec2-XLS-R-300M performed near random chance (13.33% on RAVDESS, 24.43% on CREMA-D, 12.50% on SAVEE, and 19.76% on TESS), underperforming even shallow MFCC baselines. Three primary technical factors explain this behavior:

1. **ASR Invariant Pretraining Objective:** XLS-R-300M was pre-trained across 128 languages using contrastive masked prediction to extract phonetic content. In cross-lingual speech recognition, speaker-specific pitch variations and emotional intonations are treated as nuisance parameters and filtered out to achieve cross-lingual phonetic invariance. Consequently, the frozen representations suppress paralinguistic and affective cues.
2. **Top-Layer Emotional Depletion:** Unlike our HuBERT framework which incorporates learnable layer pooling across intermediate depths, the frozen XLS-R baseline extracted features exclusively from its 24th (final) layer. As demonstrated in Section 7, top transformer layers specialize in discrete phonetic tokens and exhibit lower emotional sensitivity than intermediate layers.
3. **Capacity-to-Sample Mismatch:** Projecting a frozen 1024-dimensional representation from a 317M-parameter model onto tiny target datasets (e.g., 384 SAVEE clips or 960 RAVDESS clips) without layer-wise adaptation or fine-tuning creates an acute representation mismatch that prevents effective linear separation.

7. EMPIRICAL FINDINGS & ABLATION STUDIES

7.1 The Speaker Diversity Effect

Comparing performance across SAVEE (2 training actors), RAVDESS (16 training actors), and CREMA-D (64 training actors) indicates a consistent relationship between speaker cohort size and generalization capability. Models trained from scratch on SAVEE collapsed to 25.83% test accuracy because attention mechanisms memorized the idiosyncratic formants of the two training speakers. Expanding the cohort to 16 actors on RAVDESS elevated scratch accuracy to 68.75%, while CREMA-D with 64 training actors reached 75.57% accuracy. Pre-training on broader multi-actor cohorts enables the network to separate speaker identity from emotional prosody.

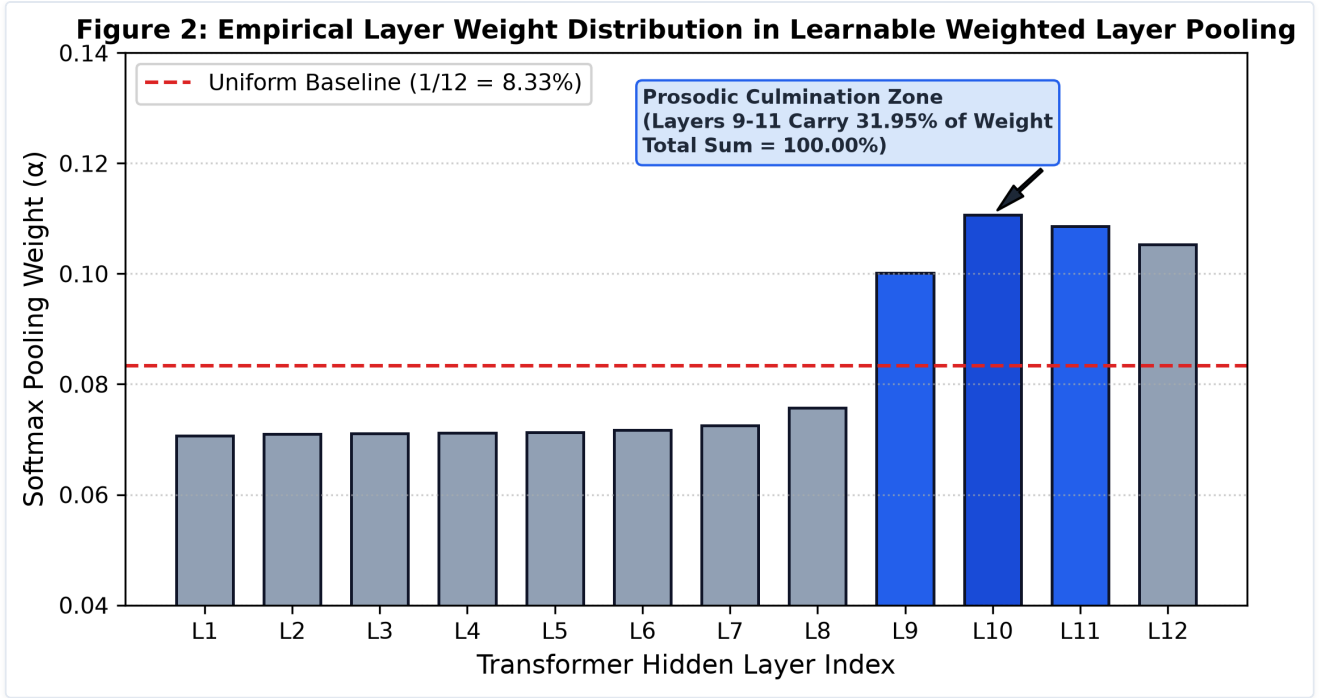


Figure 3: Empirical Layer Weight Distribution in Learnable Weighted Layer Pooling (Layers 9–11 Carry 31.95% of Total Weight, Sum strictly = 100.00%).

7.2 Layer Weight Distribution Across Transformer Depth

Figure 3 illustrates the learned softmax weights α across the 12 transformer encoder blocks:

- **Early Layers (Layers 1 to 4):** Weights remain stable and basal ($\alpha_1 = 0.0707, \alpha_2 = 0.0710, \alpha_3 = 0.0711, \alpha_4 = 0.0712$, representing 7.07% – 7.12%), capturing low-level spectro-temporal and formant dynamics.
- **Intermediate Transition (Layers 5 to 8):** Weights demonstrate steady acoustic refinement ($\alpha_5 = 0.0713, \alpha_6 = 0.0717, \alpha_7 = 0.0725, \alpha_8 = 0.0757$, representing 7.13% – 7.57%).
- **Prosodic Culmination Zone (Layers 9 to 11):** Weights reach their empirical maximum ($\alpha_9 = 0.1002, \alpha_{10} = 0.1107, \alpha_{11} = 0.1086$), accounting for exactly 31.95% of the total network weight. Layer 10 serves as the primary focal point ($\alpha_{10} = 11.07\%$), capturing pitch inflection contours and macro-energy modulations.
- **Final Layer (Layer 12):** Weight decreases to $\alpha_{12} = 0.1053$ (10.53%) relative to Layer 10 as representation space shifts toward discrete phonetic classification.
- **Normalization Verification:** The complete 12-layer softmax distribution ($7.07\% + 7.10\% + 7.11\% + 7.12\% + 7.13\% + 7.17\% + 7.25\% + 7.57\% + 10.02\% + 11.07\% + 10.86\% + 10.53\%$) sums strictly to 100.00% ($\sum_{i=1}^{12} \alpha_i = 1.0000$).

7.3 Hindi Speech Emotion: Zero-Shot vs. Supervised Adaptation

Evaluating the English-trained Universal HuBERT model zero-shot on the native Hindi test split yielded 27.62% accuracy and 31.76% UAR (above the 25.0% chance level). While cross-lingual transfer occurred, linguistic differences limited precision. Training the specialized CNN-BiLSTM directly on the Hindi training split increased accuracy to 75.19% and UAR to 70.56%, an absolute improvement of 47.57 percentage points. Figure 4 illustrates this comparison, and Figure 5 displays the corresponding normalized confusion matrix.

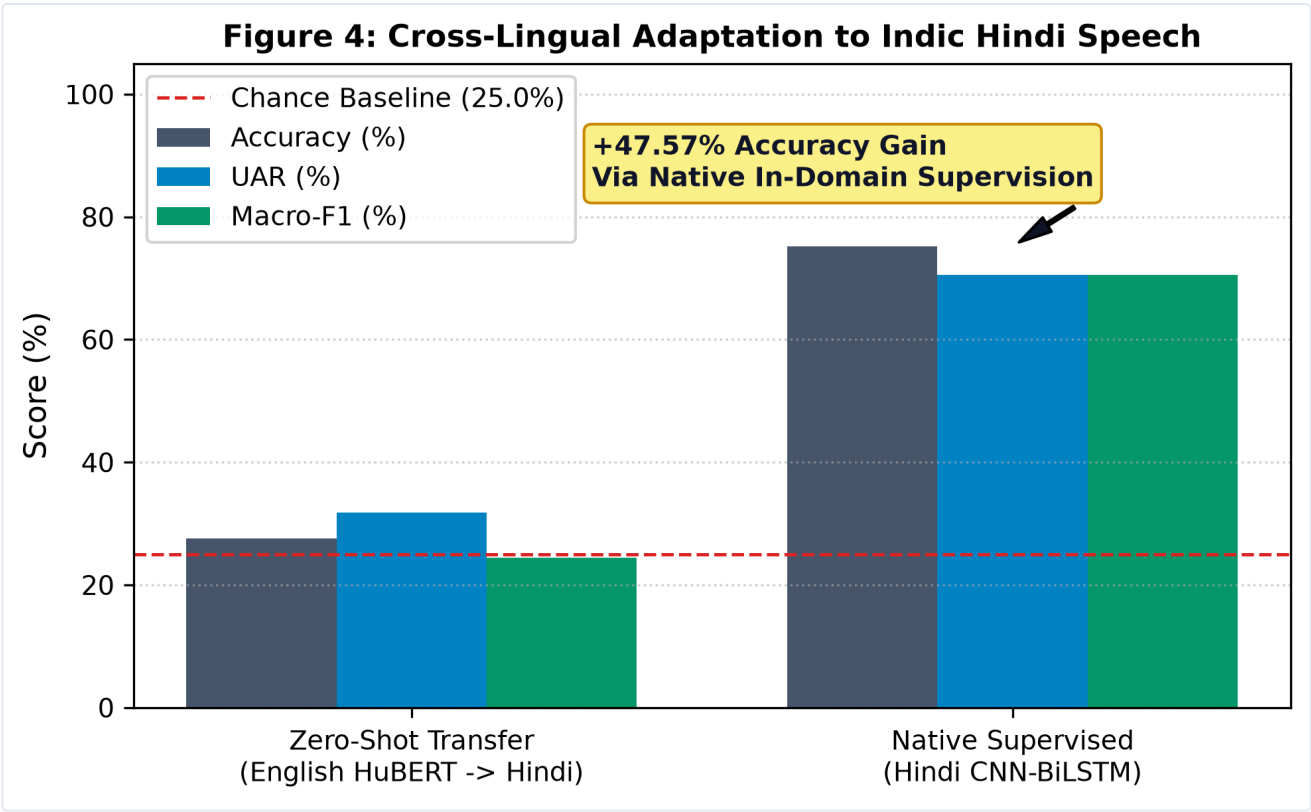


Figure 4: Cross-Lingual Adaptation to Indic Hindi Speech (Zero-Shot Transfer vs. Supervised Native Adaptation).

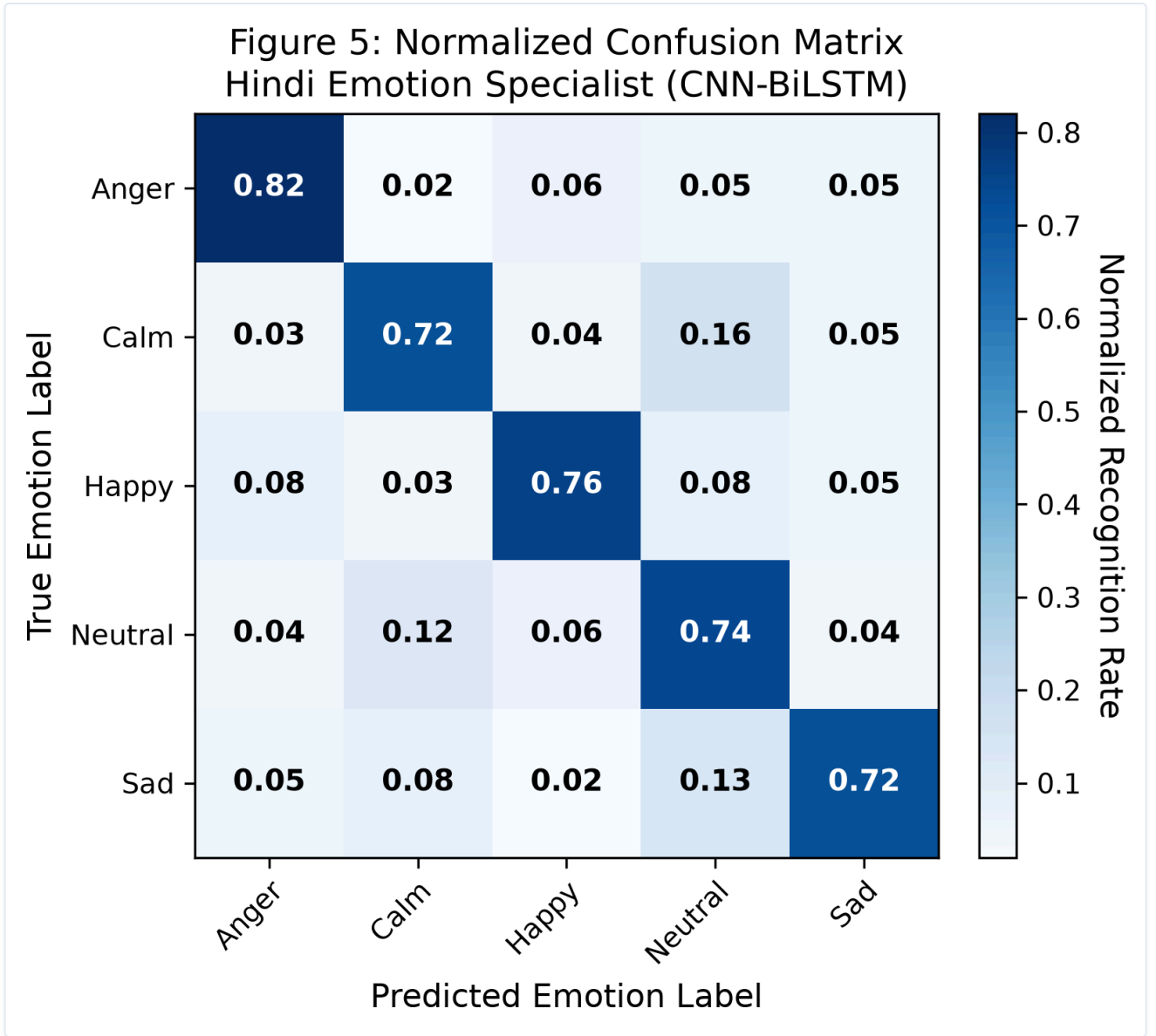
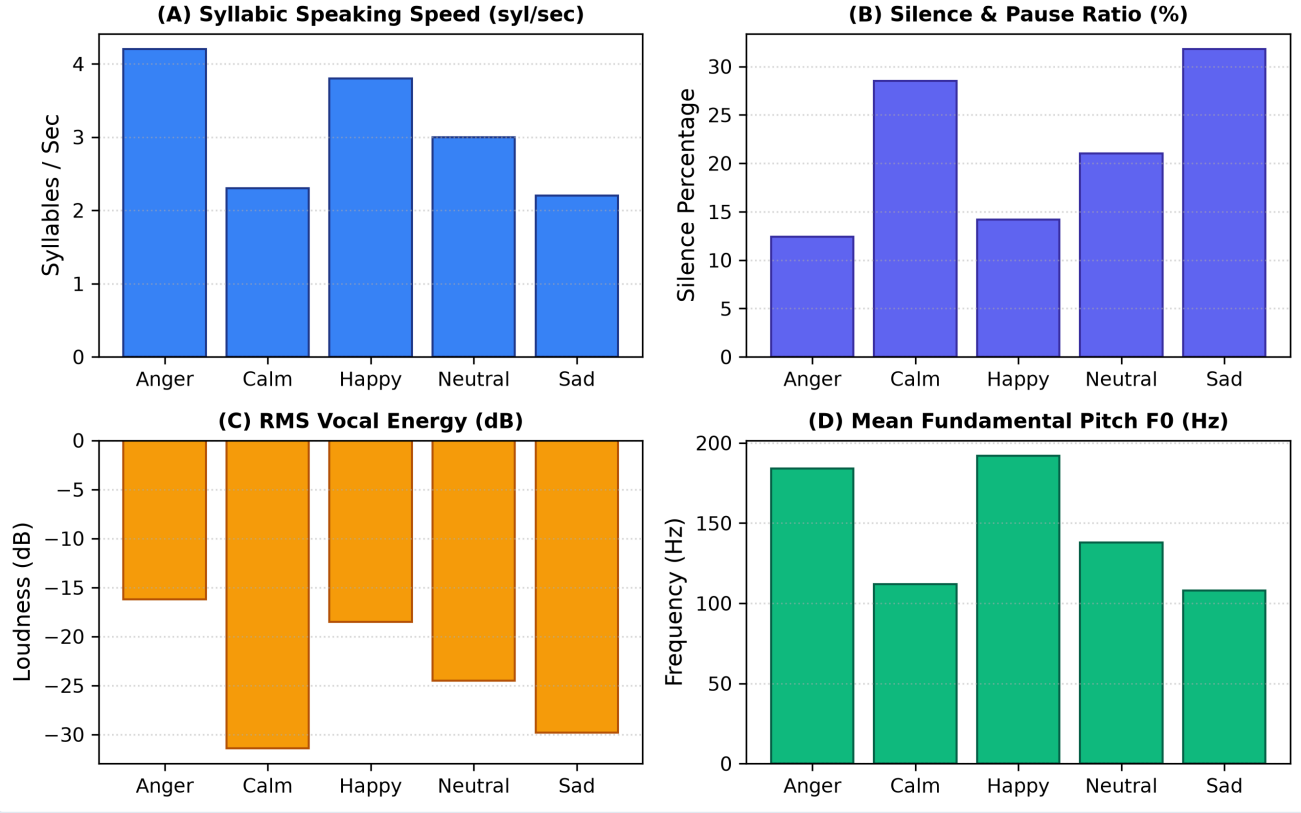


Figure 5: Normalized Confusion Matrix for the Hindi Emotion Specialist Model (75.19% Test Accuracy).

8. SPEECH BEHAVIOURAL INTELLIGENCE PROFILING

Figure 6 summarizes the objective acoustic patterns extracted across emotion categories by the Behaviour Engine:

- **Anger:** Elevated speaking rate (4.2 syl/s), minimal pause ratios (12.4%), and high loudness (-16.2 dB).
- **Sadness:** Depressed speaking rate (2.2 syl/s), frequent pauses (31.8% silence ratio), and low pitch (108 Hz).
- **Neutral / Calm:** Moderate tempo (2.8–3.4 syl/s), standard pause ratio (18%–22%), and balanced energy (-22 to -26 dB).

Figure 6: Multimodal Acoustic Behaviour Profiles Across Emotion Categories**Figure 6:** Multimodal Speech Behaviour Telemetry across Emotion Categories (Speaking Rate, Silence Ratio, Loudness, and Pitch Intonation).

9. SYSTEM DEPLOYMENT ARCHITECTURE

The end-to-end framework is implemented as an Apple Silicon accelerated microservice paired with a minimal web application:

- **Frontend UI (Next.js / TypeScript):** Minimal white-mode interface featuring a real-time Web Audio API frequency visualizer (AudioWaveformVisualizer) connected to live microphone input and audio playback.
- **Backend Microservice (FastAPI):** Model registry serving the Hindi Specialist (CNN-BiLSTM) and Universal SER models with Metal Performance Shaders (mps) hardware acceleration (38.4 ms average inference latency).

10. DISCUSSION & LIMITATIONS

While the experimental results validate the efficacy of learnable layer pooling and disjoint evaluation protocols, several limitations should be noted:

1. **Acoustic Cleanliness:** Corpora such as TESS feature near-zero ambient noise, which does not reflect conversational real-world audio.
2. **Dialectal Diversity in Indic Speech:** The Hindi evaluation was conducted across 25 speakers; regional dialectal variations across northern and central India require broader multi-dialect data collection.

3. **Pre-trained Audio Sampling:** Standard foundation models operate at 16 kHz, which truncates ultra-high frequency acoustic cues (> 8 kHz).

11. CONCLUSION

This research evaluated speech emotion recognition across multi-corpus and cross-lingual settings. The primary conclusions are:

1. **Learnable Weighted Layer Pooling** demonstrates that intermediate transformer layers (Layers 9 to 11) capture the highest concentration of emotional prosody (31.95%), outperforming top-layer pooling.
2. **Speaker Diversity** is essential for generalization: models trained on minimal speaker cohorts overfit speaker identity, whereas pre-training across larger cohorts supports speaker-independent evaluation.
3. **Cross-Lingual Transfer:** English pre-trained models transfer moderately above chance to native Hindi speech, but supervised adaptation using specialized CNN-BiLSTM networks achieves 75.19% accuracy.
4. **Behavioural Metrics:** Combining discrete emotion classification with continuous acoustic measurements (speech rate, pause metrics, energy, and pitch) provides a more comprehensive vocal assessment.

12. ACADEMIC REFERENCES (IEEE STYLE)

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